



Winding Tree

Travel Industry's Invisible Battle



Maksim Izmaylov

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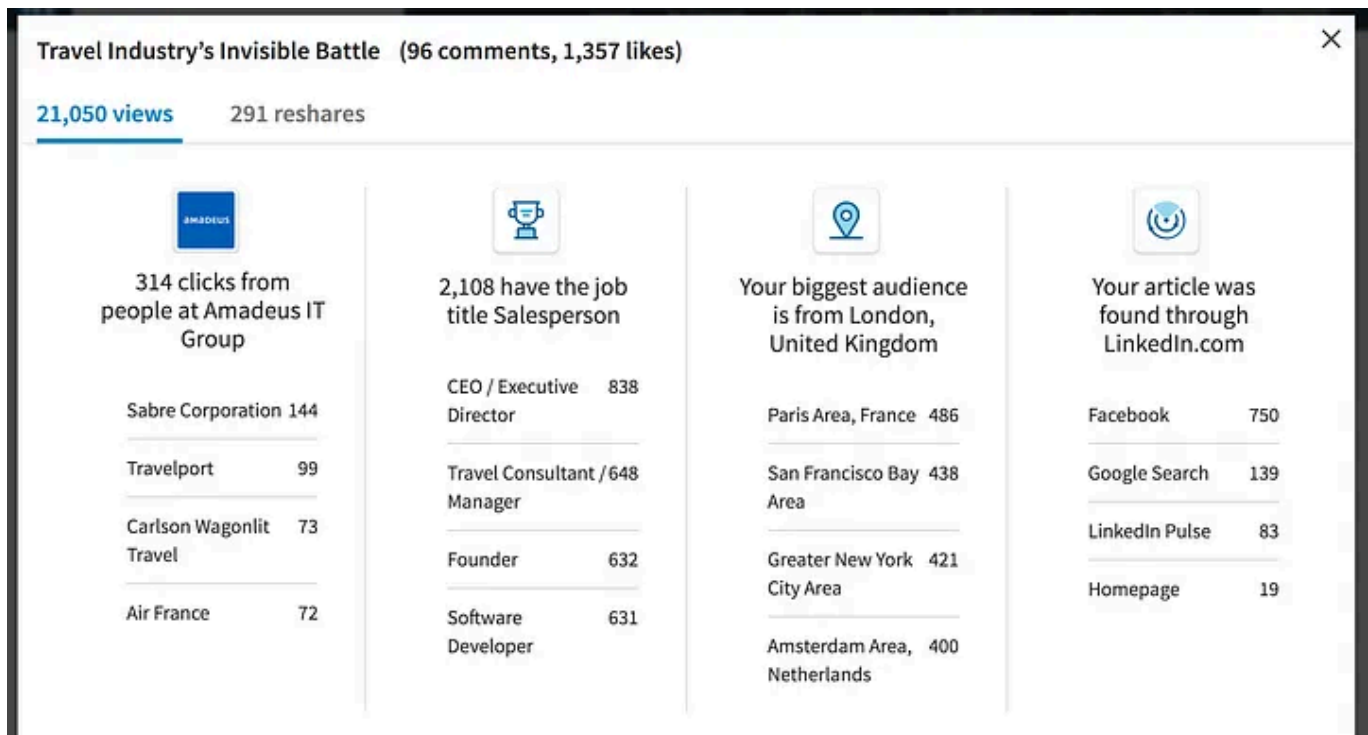
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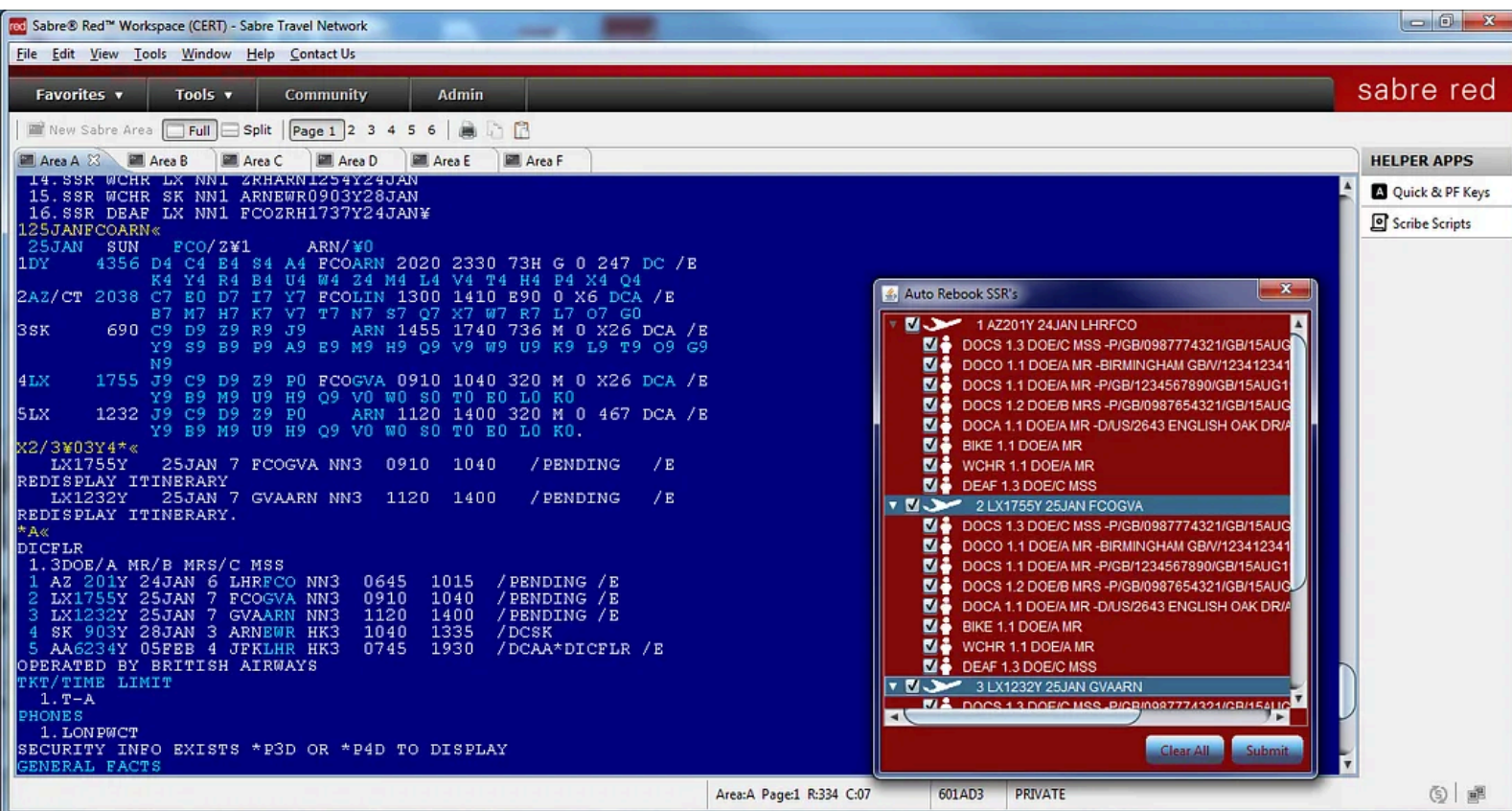


Note From Author, September 24th, 2017



This article was originally posted on LinkedIn, where it managed to attain quite a readership (see the stats). I'd like to thank everyone commented on it, shared it and read it. Thank you, these numbers are mind-blowing to me!

The great news today is that the article was published in Russian by the team behind Travel Tech Con Russia (not affiliated with the conference I started in San Francisco — Travel Tech Con).



Sabre Red Interface, Not Long Ago

For an outsider the travel industry seems to be doing better than ever. These days you can find any type of accommodation, mode of transportation or an activity to your liking and book it online. The problems is that we visit thirty eight websites, on average, before booking travel. Well, it's not the problem, it's just a symptom of a series of underlying problems.

For someone who knows a thing or two about travel, it might seem that travel companies, including big OTAs, hotels and airlines, are not doing enough to improve customer experience and instead they are organizing their services around their businesses' needs.

Note. OTA stands for online travel agency. Two largest OTAs in the world are Priceline and Expedia, but any website or app that sells travel can be referred to as

an OTA. GDS stands for global distribution system, with three major players being Amadeus, Sabre and Travelport.

Why, for example, are hotels and airlines bombarding you with commercials that aim to convince you to book directly on their websites? Because they want you to have better experience? Not necessarily. On their own websites they are able to sell you additional services, without paying commission fees to OTAs and GDSs. And sometimes even on their own turf they fail quite miserably.

In a recent experience, I tried to sign up for the Flying Blue program (joint project of Air France, Delta and KLM) and was unable to do so, apparently due to a technical error, that lasted for three consecutive days. I ended up calling their customer support line. You've read it right; I actually had to speak to a human being to be able to sign up. The task that should take at most five minutes, took me a few hours to complete. Here's a fun question: how many airlines does it take to fix a website registration form?

Another symptom of deep underlying problems is that no one talks about the bad stuff, the giant elephant in the room. Whenever I attend a travel conference, everything seems to be just great: each airline and airport now have apps, they're on track of delivering their own chatbots and VR experience. And yes, there is the video from Google that you've seen a few times at other conferences already. It doesn't actually say anything, but — look! — there are kids playing with Google Cardboards!

Baggage loss, flight cancellations, hotel overbookings, all that happens on regular basis and these are the experiences that change customer behavior drastically. Have you ever heard someone saying “I’d not prefer flying with [airline name] because they don’t have [movie name] on their entertainment systems”. But so many times I’ve heard statements like “I no longer check baggage after [airline name] had lost it once”.

Take a look at #paxes hashtag (paxex stands for passenger experience), it seems to be all about new aircraft seats, miles and upgrades, but I’d like to argue that passenger experience is so much bigger than that.

Real stories of passenger experience are not about the room in between the seats or new aircraft liveries, the real stories are filled with desperation, frustration and anger and often accompanied by #neveragain hashtag.

In this article I’d like to point out the root cause of these problems and suggest a solution that can greatly improve customer experience, enable innovation, and help suppliers improve their bottom lines.

Uberization of Travel

It’s a fact of life that airlines and hotels have already lost control of some aspects of their businesses and that may explain the fact that they only improve customer experience in their domains, i.e. actual aircrafts and hotels.

In a recent conversation with an executive from a major international airline I learned that many airlines these days can't even price their products. All they do is give their inventory to one of the three GDSs that together control the majority of the market worldwide. While it's hard to estimate their combined market share, according to various sources it might be close to 99%.

The situation is very similar in hospitality. Priceline and Expedia together control 93% of non-direct hotel bookings in the US (source); all the OTA brands that you know belong to one of those two companies, e.g. Hotels.com is Expedia, Booking.com is Priceline. Globally these two companies accounted for 47% of the market in 2013, up from 38% in 2011 (source).

92% of all hotels bookings in the US go through websites that belong to one of the two largest OTAs: Priceline and Expedia.

Hotels don't control the whole customer journey anymore and neither do airlines. What's happening is *uberization* of those industries and the new reality in the age of Uber is very grim. According to some sources, many drivers working for Uber don't make minimum wage, and that's even before we take depreciation of their assets into account.

In a few years, airlines and hotels will, theoretically, only manage the physical assets that they own, just like Uber drivers, and companies that would manage their inventory for them will make the most money. And it's incredibly important to see how dangerous this situation is. Whole industries give up crucial parts of their businesses to a few gigantic corporations. Why is this happening? What are other implications of this situation and how do we fix it?

The Problem: Monopolies

The problem is that a few years ago airlines and hotels decided that they are not good with technology anyway, so airlines created GDSs and hotels gave their inventory to OTAs. No one planned for long term. If they started with creation of open standards for data exchange, we wouldn't have this situation on our hands. Switching between providers would have been easy and we'd have prevented the creation of those monopolies.

Now these companies can easily threaten their customers, for example, Sabre bullied US Airways into signing a contract on Sabre's terms by threatening the airline to cut them off from their system completely. The airline later sued and won the case. One of the arguments in court was that, allegedly, Sabre conspired with two other GDSs, Amadeus and Travelport, and while there was no evidence to support it, in the last few years Sabre settled a few similar claims from other airlines.

The situation is as complicated in the hospitality sector. Expedia, for example, has been accused of buying advertising for properties not listed on their website and offering interested customers one of their partner hotels instead. In the recent years Expedia has also been accused of charging its customers more taxes than they'd otherwise pay, and at the same time a number of local US governments claimed that Expedia didn't pay enough taxes. At the same time Expedia and Priceline spend billions of dollars on advertising.

Hotels and airlines are trying to break free from monopolies. For example, at the end of 2015, Lufthansa introduced a €16 surcharge fee for tickets that are booked through GDSs.

The hotel industry reacted to these problems by creating Roomkey, a joint project of six major hotel brands. Roomkey was designed to be a replacement for big OTAs, but the project was not successful.

No Innovation in Travel

The existence of these monopolies is a problem for everyone. It's a problem for travel suppliers, as I've shown above, it's also a problem for travel startups and that's why we haven't seen any new travel startups innovating core travel products, i.e. actual airline and hotel bookings. Out of twenty-something travel startups presenting at the last Plug And Play Winter Summit there was only one doing something about hotel bookings.

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Signup Rejected
To: [REDACTED]
Dear Sir or Madam,
Thank you for your registration and interest in joining [Booking.com](#) affiliate program.
We have reviewed your website and unfortunately we do not feel that [REDACTED] has some of the fundamental elements we feel would make for a mutually successful collaboration.
It is important for us to see a well structured website with rich unique content that contains both pictures and text and is preferably travel related.
Moreover, we would like to see qualified web traffic and at the moment [REDACTED] does not meet this criteria.
Again, thank you for contacting us and feel free to contact us in the near future should [REDACTED] come to meet the qualities outlined above.
With kind regards,
[Booking.com](#) distribution team

HotelsCombined Affiliate Support
Re: Enquiry from [REDACTED] Data feeds access
To: [REDACTED]
Hi [REDACTED]
To clarify, our hotel data feeds consist of a static list with basic hotel information. Access is offered on a limited basis, depending on existing partner site content as well as account performance. In addition, our hotel data feeds are not a suitable solution for new websites built from scratch and with no/little existing content.
Please tell us more about your website and send us some examples on how you would like to integrate our hotel data with your existing content, so that we can look into your request.
Regards
[REDACTED]
 HotelsCombined   

Are you a startup? No data for you!

I've met many travel startups and individual hackers through my work at Travel Tech Con, a community for travel startups. I always ask those entrepreneurs the same question: what is the hardest problem that you are facing? The answer is, invariably, the same: access to hotel/airline inventory.

I can definitely relate to that, because we had the same problem at Roomstorm. We needed access to hotel inventory, and integrations that we'd completed took weeks, if not months, and we wouldn't have been able to sway our current partners into working with us if we didn't have substantial funding and Y Combinator brand behind us. The reality is, if you are an independent entrepreneur or an early stage startup, it will be extremely hard for you to obtain that data.

The only way to innovate is to try different things quickly. "Fail fast" mantra is at the heart of Silicon Valley, but the travel industry, dominated by a few large corporations, is purposely preventing startups from applying it.

Customer Experience

What does it mean for customers? It simply means that we overpay for travel products and services and a few middlemen collect the rent. Combined valuation of the three major GDSs is almost \$30 billion dollars. The two aforementioned OTAs account for staggering \$90 billion of value that came from us, travelers. These companies add 10–20% to the cost of travel. Do their services priced fairly? It's hard to say, but let's dig a bit deeper.

First of all, as a airline customer, for example, you must know that the IT solutions that GDSs provide to airlines are 10–20 years old and system-wide airline outages due to failures of those systems happen several times a year.

If at the airport you've noticed that the airline agent is typing on the keyboard a bit too much for the task, they're probably using software that looks like this.

Sabre Red: Typical "Green" Screen Airline Software

The software that looks like this requires special hardware to run it — mainframes. Many argued that mainframes can be the single point of failure, but we will never know real reasons behind system-wide airline outages. What is known for sure, though, is that these system are incredibly costly to procure, maintain and upgrade. They also lack an appropriate level

of security, as it has been proved by security experts Karsten Nohl and Nemanja Nikodijevic:

Airline reservation systems grew from mainframes with green-screen terminals to modern-looking XML/SOAP APIs to access those same mainframes.

The systems lack central concepts of IT security, in particular good authentication and proper access control.

We show how these weaknesses translate into disclosure of traveler's personal information and would allow several forms of fraud and theft, if left unfixed.

Not only these systems are costly and error prone, they are also incredibly hard to update. We can safely speculate that software that runs on those mainframes is updated several times a year at best, as opposed to modern software systems, like Facebook, Google or any startup in Silicon Valley that updates their software multiple times per day.

Hotels also have a troubled relationship with OTAs, as I described above, which results in excessive fees and unsatisfactory customer experience. The source of these problems is, by and large, the same: outdated systems, lack of competition and therefore no incentive to update them, customer lock-down, etc.

Airlines to Blame

In many cases the problems that we, as consumers, experience with airlines are inflicted by airlines themselves. A study from 2015 (PDF) shows that US travelers pay additional \$6.7B in costs due to limited ability to compare prices on different airlines.

In this light attempts of airlines and hotels to bring customers to their websites can be explained very simply: they want to limit your ability to compare prices, because they will be able to sell you the same service for a higher price.

Shaping The Future of Travel

Where do we go from here and how do we get there? Here's a few ideas.

Acceptance Test for Travel

Easy access to data is key to innovation, so in the ideal scenario, travel inventory data will be easily accessible via a set of APIs on an open and transparent marketplace.

If an individual software engineer is able to access hotel, airline and tour-and-activity data easily, combine them in an interesting way and iterate quickly, then the test is passed.

Not until people can play around with information does it becomes truly accessible.

— Ariel Waldman, spacehack.org

Innovation Happens Bottom-Up

Software engineers working in the travel industry should be able to work together on common problems. Currently there is very little collaboration and no open-source projects in travel.

Standardization and Inclusion

There were a few attempts to create standards for the travel industry. Unfortunately all of them are very exclusive, developed behind closed doors. A recent example is the NDC standard from IATA, which took at least three years to release. Travel is already ten years behind and by artificially limiting collaboration we'll just enlarge that gap.

Decentralization

Standards would certainly help to prevent creation of monopolies, but the travel industry is ready for the upgrade to the next level. The blockchain technology can solve all the problems I outlined above. It's still in a very

early stage, but it will radically transform many business models that we know today. Here is a great explanation of how decentralized Uber or AirBnB could work, and please note the analogy with the businesses of aforementioned OTAs and GDSs:

Skip to 9:20, if you're in a rush

Temporary monopolies are a great way to create wealth, but it seems like in the next few years a completely new type of businesses will emerge and here's my version of a blockchain-enabled travel distribution business.

Solution: Decentralized Marketplace for Travel

The solution that I propose is a radical one, but the travel industry is beyond ripeness for this upgrade. It will democratize access to travel data and prevent the existence of monopolies. It's in the works and its name is Winding Tree.

The blockchain technology allows us to build decentralized organizations. It means that we can create a single company that will solve many problems mentioned above, but at the same time it will not have any downsides of a monopoly.

To better understand how blockchain works please watch the video above and read the article where [I outline the most important aspects of this technology.](#)

Here's how it works. Winding Tree as an organization doesn't have any employees. Instead all the business logic that is expressed in computer code, which is deployed on a public blockchain, like Ethereum. The fact that we are using public blockchain allows everyone to be sure that the platform creators will not be able to artificially limit someone's access to the platform (Sabre vs US Airways) or change its business model, e.g. start charging transaction fees.

Open Marketplace

Winding Tree is an open marketplace, which means that the barriers to entry are very low. Both buyers and suppliers will be able to integrate with it via simple APIs.

Full Automation

Winding Tree has no employees running its operations, it's fully automated. It means that there will be no human errors. Our goal is to integrate with hotel and airline systems directly in order to access real-time availability.

No Transaction Fees

We believe that charging a transaction fee is an outdated business model. The marketplace will be automatically calculating fees for making API calls to the system, which will be in range of one penny per call.

Payment Reconciliation

Winding Tree will have a default payment reconciliation solution with fees in the range of less than 1%. Users of the platform will be able to easily switch to the payment provider of their choice, if they want to.

Open Collaboration

Winding Tree source code will be open-sourced when we release an MVP and we will encourage and welcome software engineers to build applications and businesses on top of our platform.

Proprietary Extensions

One of the ways for us as a company to make money is development of proprietary extensions on top of the Winding Tree platform. These extensions will range from pricing algorithms for suppliers to insurance products, etc.

100% Uptime

Winding Tree MVP will be based on Ethereum, a public blockchain, which will allow it to never go down.

Conclusion

Our goal with Winding Tree is to enable innovation and collaboration in the travel industry. We can change the whole industry, but we won't be able to do it without your help and feedback, which you can give on our [slack channel](#), by joining our mailing list on our website windingtree.com, or simply via email: join@windingtree.com.

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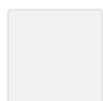
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Dec 27, 2017



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So: how... [more](#)



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Flyign Fish the Incredible One

Dec 27, 2017



My oh my...

If I may play the devils advocate here (as no one really seems to respond):

A lot of one-sided views here have you ever thought about the fact that the large-scale OTAs are also the advocates of their clients against hotels? Blockchain is... [more](#)



[Reply](#)



Emin Sharifov

Feb 24, 2017



Great Article [Maksim Izmaylov](#), but the article is targeting only the booking process. Good chunk of an OTA's value and revenue comes from post-booking processes. Meanwhile Airlines are also competing for these extra convenience fees but are having... [more](#)



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